KAIZEN MATHS

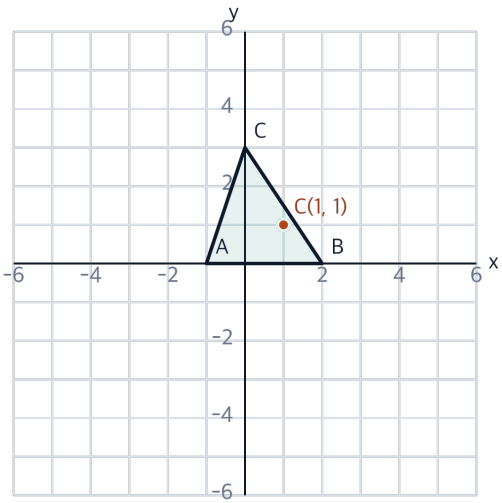
Lesson starter

Transformations

Challenge Transformations

Name _____

SA1



Enlarge shape ABC by scale factor -2 from centre (1, 1). Give the coordinates of the image.

GEO-C8E6-L4 · SA

Show your working

KAIZEN MATHS

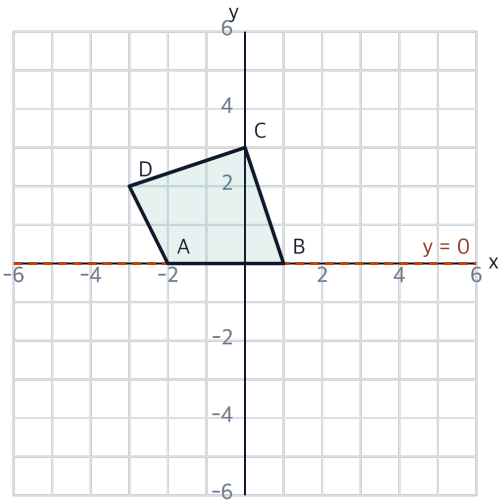
Lesson starter

Transformations

Challenge Transformations

Name _____

SB1



First translate shape ABCD by $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$, then reflect the result in the line y = 0. Give the final image coordinates.

GEO-C8E6-L4 · SB

Show your working

KAIZEN MATHS

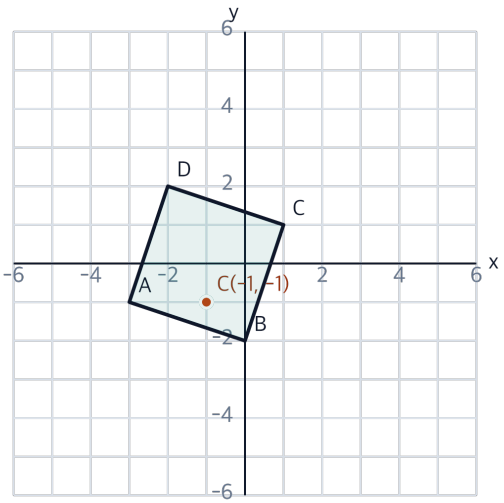
Lesson starter

Transformations

Challenge Transformations

Name _____

SC1



Enlarge shape ABCD by scale factor -1 from centre (-1, -1). Give the coordinates of the image.

GEO-C8E6-L4 · SC

Show your working

KAIZEN MATHS

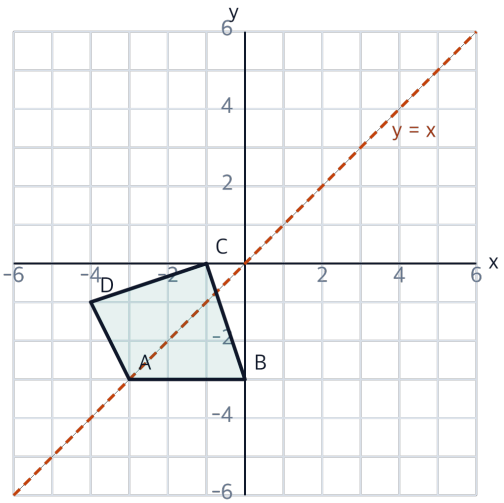
Lesson starter

Transformations

Challenge Transformations

Name _____

SD1



First translate shape ABCD by $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$, then reflect the result in the line y = x. Give the final image coordinates.

GEO-C8E6-L4 · SD

Show your working

KAIZEN MATHS

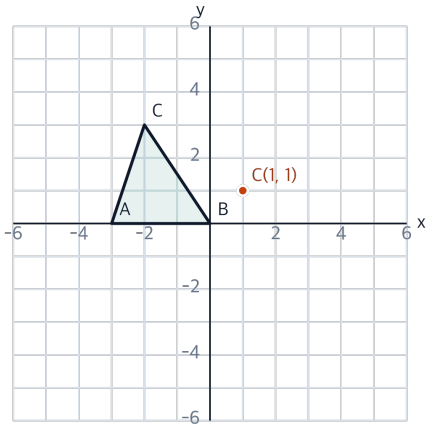
Exit ticket

Transformations

Challenge Transformations

Name _____

EA1



Enlarge shape ABC by scale factor -1 from centre (1, 1). Give the coordinates of the image.

Show your working

KAIZEN MATHS

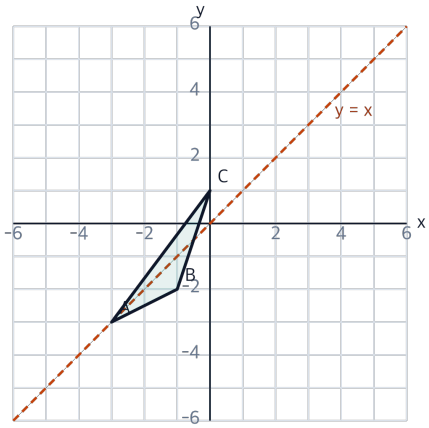
Exit ticket

Transformations

Challenge Transformations

Name _____

EB1



First translate shape ABC by $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$, then reflect the result in the line y = x. Give the final image coordinates.

Show your working

KAIZEN MATHS

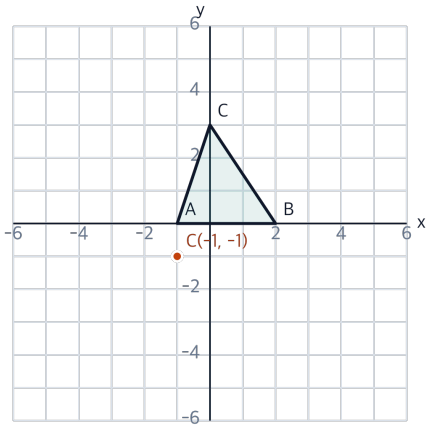
Exit ticket

Transformations

Challenge Transformations

Name _____

EC1



Enlarge shape ABC by scale factor -1 from centre (-1, -1). Give the coordinates of the image.

Show your working

KAIZEN MATHS

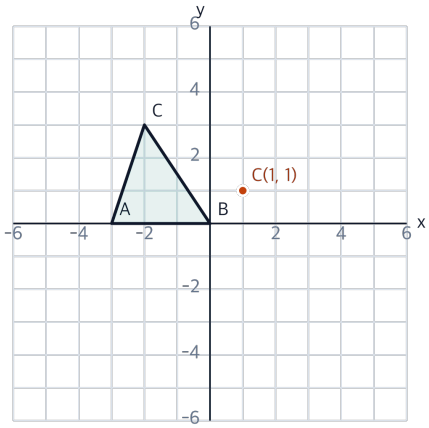
Exit ticket

Transformations

Challenge Transformations

Name _____

EA1



Enlarge shape ABC by scale factor -1 from centre (1, 1). Give the coordinates of the image.

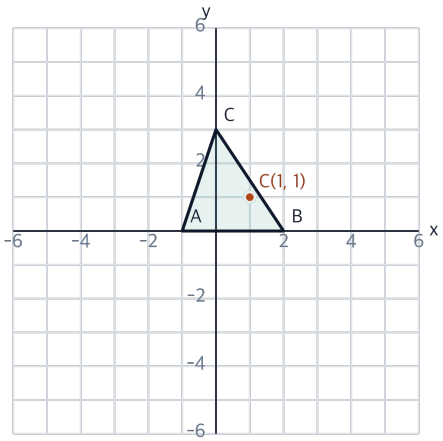
Show your working

Teacher answers & working

Transformations · Challenge Transformations

GEO-C8E6-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Enlarge shape ABC by scale factor -2 from centre (1, 1). Give the coordinates of the image.

Answer:

A′(5, 3), B′(-1, 3), C′(3, -3)

Working:

A negative scale factor sends each point through the centre of enlargement.

Use centre (1, 1) and multiply each centre-to-vertex vector by -2.

Vertex	Original	Vector from centre	Scaled vector	Image
A	(-1, 0)	(-2, -1)	-2(-2, -1) = (4, 2)	A′ (5, 3)
B	(2, 0)	(1, -1)	-2(1, -1) = (-2, 2)	B′ (-1, 3)
C	(0, 3)	(-1, 2)	-2(-1, 2) = (2, -4)	C′ (3, -3)

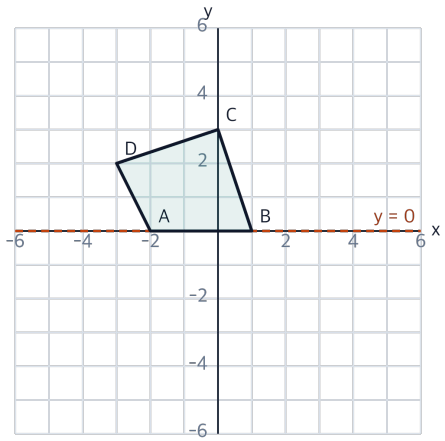
Final image coordinates: A′(5, 3), B′(-1, 3), C′(3, -3).

Teacher answers & working

Transformations · Challenge Transformations

GEO-C8E6-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



First translate shape ABCD by $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$, then reflect the result in the line y = 0. Give the final image coordinates.

Answer:

A′(0, 2), B′(3, 2), C′(2, -1), D′(-1, 0)

Working:

Apply the transformations in the order written.

Step 1: translate every vertex by $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$.

Step 2: reflect the translated points in y = 0.

Vertex	Start	After step 1	Final image
A	(-2, 0)	translate by (2, -2): (0, -2)	reflect in y = 0: A' (0, 2)
B	(1, 0)	translate by (2, -2): (3, -2)	reflect in y = 0: B' (3, 2)
C	(0, 3)	translate by (2, -2): (2, 1)	reflect in y = 0: C' (2, -1)
D	(-3, 2)	translate by (2, -2): (-1, 0)	reflect in y = 0: D' (-1, 0)

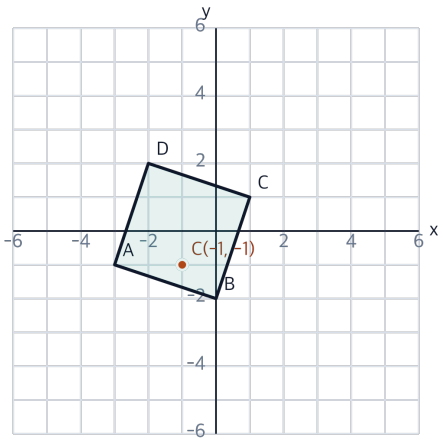
Final image coordinates: A'(0, 2), B'(3, 2), C'(2, -1), D'(-1, 0).

Teacher answers & working

Transformations · Challenge Transformations

GEO-C8E6-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Enlarge shape ABCD by scale factor -1 from centre (-1, -1). Give the coordinates of the image.

Answer:

A'(1, -1), B'(-2, 0), C'(-3, -3), D'(0, -4)

Working:

A negative scale factor sends each point through the centre of enlargement.

Use centre (-1, -1) and multiply each centre-to-vertex vector by -1.

Vertex	Original	Vector from centre	Scaled vector	Image
A	(-3, -1)	(-2, 0)	-1(-2, 0) = (2, 0)	A' (1, -1)
B	(0, -2)	(1, -1)	-1(1, -1) = (-1, 1)	B' (-2, 0)
C	(1, 1)	(2, 2)	-1(2, 2) = (-2, -2)	C' (-3, -3)
D	(-2, 2)	(-1, 3)	-1(-1, 3) = (1, -3)	D' (0, -4)

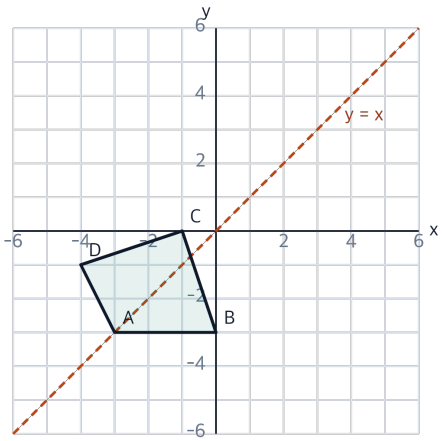
Final image coordinates: A'(1, -1), B'(-2, 0), C'(-3, -3), D'(0, -4).

Teacher answers & working

Transformations · Challenge Transformations

GEO-C8E6-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



First translate shape ABCD by $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$, then reflect the result in the line y = x. Give the final image coordinates.

Answer:

A′(0, -5), B′(0, -2), C′(3, -3), D′(2, -6)

Working:

Apply the transformations in the order written.

Step 1: translate every vertex by $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

Step 2: reflect the translated points in y = x.

Vertex	Start	After step 1	Final image
A	(-3, -3)	translate by (-2, 3): (-5, 0)	reflect in y = x: A′ (0, -5)
B	(0, -3)	translate by (-2, 3): (-2, 0)	reflect in y = x: B′ (0, -2)
C	(-1, 0)	translate by (-2, 3): (-3, 3)	reflect in y = x: C′ (3, -3)
D	(-4, -1)	translate by (-2, 3): (-6, 2)	reflect in y = x: D′ (2, -6)

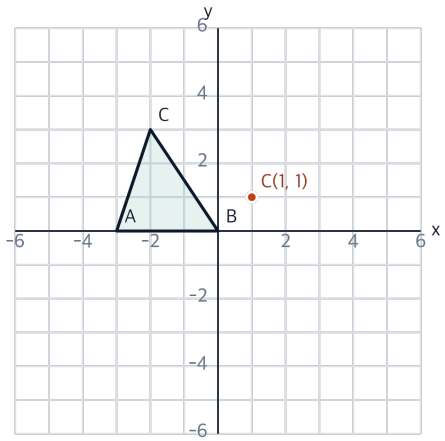
Final image coordinates: A′(0, -5), B′(0, -2), C′(3, -3), D′(2, -6).

Teacher answers & working

Transformations · Challenge Transformations

GEO-C8E6-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Enlarge shape ABC by scale factor -1 from centre (1, 1). Give the coordinates of the image.

Answer:

A'(5, 2), B'(2, 2), C'(4, -1)

Working:

A negative scale factor sends each point through the centre of enlargement.

Use centre (1, 1) and multiply each centre-to-vertex vector by -1.

Vertex	Original	Vector from centre	Scaled vector	Image
A	(-3, 0)	(-4, -1)	-1(-4, -1) = (4, 1)	A' (5, 2)
B	(0, 0)	(-1, -1)	-1(-1, -1) = (1, 1)	B' (2, 2)
C	(-2, 3)	(-3, 2)	-1(-3, 2) = (3, -2)	C' (4, -1)

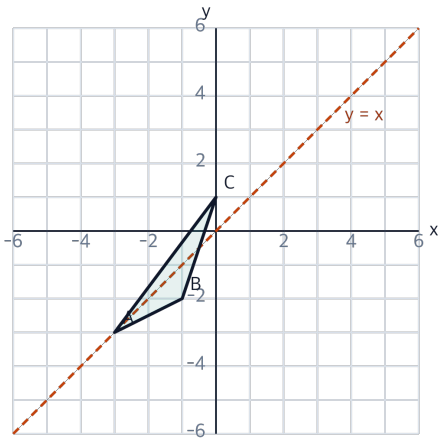
Final image coordinates: A'(5, 2), B'(2, 2), C'(4, -1).

Teacher answers & working

Transformations · Challenge Transformations

GEO-C8E6-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



First translate shape ABC by $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$, then reflect the result in the line y = x. Give the final image coordinates.

Answer:

A′(-1, 0), B′(0, 2), C′(3, 3)

Working:

Apply the transformations in the order written.

Step 1: translate every vertex by $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$.

Step 2: reflect the translated points in y = x.

Vertex	Start	After step 1	Final image
A	(-3, -3)	translate by (3, 2): (0, -1)	reflect in y = x: A′ (-1, 0)
B	(-1, -2)	translate by (3, 2): (2, 0)	reflect in y = x: B′ (0, 2)
C	(0, 1)	translate by (3, 2): (3, 3)	reflect in y = x: C′ (3, 3)

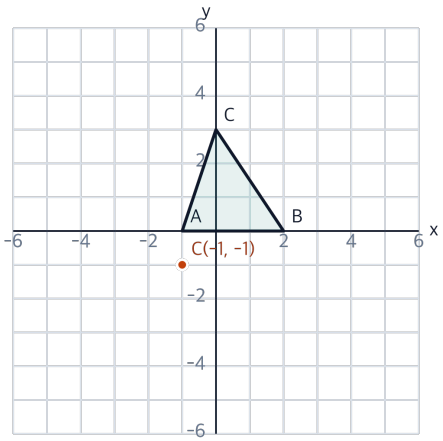
Final image coordinates: A′(-1, 0), B′(0, 2), C′(3, 3).

Teacher answers & working

Transformations · Challenge Transformations

GEO-C8E6-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Enlarge shape ABC by scale factor -1 from centre (-1, -1). Give the coordinates of the image.

Answer:

A'(-1, -2), B'(-4, -2), C'(-2, -5)

Working:

A negative scale factor sends each point through the centre of enlargement.

Use centre (-1, -1) and multiply each centre-to-vertex vector by -1.

Vertex	Original	Vector from centre	Scaled vector	Image
A	(-1, 0)	(0, 1)	-1(0, 1) = (0, -1)	A' (-1, -2)
B	(2, 0)	(3, 1)	-1(3, 1) = (-3, -1)	B' (-4, -2)
C	(0, 3)	(1, 4)	-1(1, 4) = (-1, -4)	C' (-2, -5)

Final image coordinates: A'(-1, -2), B'(-4, -2), C'(-2, -5).